

Quantum Reservoir Computing for Realised Volatility Forecasting

Team Artemis | Global Industry Challenge 2026 | Phase 2 | Track A: Financial Volatility

1. Track Selection and Problem Framing

We target **Track A**. The sub-problem is one-month-ahead forecasting of *realised volatility* (RV) of the S&P 500, with particular attention to the calm-to-turbulent regime transitions that drive risk-capital allocation and derivatives pricing. RV is an ideal match for reservoir computing for three structural reasons. First, RV exhibits strong *long memory*: shocks decay slowly and hyperbolically, which a reservoir with fading memory can store without an explicit recurrence to train. Second, it is *multi-scale*, with daily, weekly and monthly components (the empirical basis of the HAR model), so a reservoir read out at several evolution times can expose features at matched scales. Third, regime transitions are sharply *nonlinear*, and a quantum reservoir supplies a high-dimensional nonlinear feature map at no training cost. Because reservoir computing fixes the dynamical system and trains only a linear readout, it is gradient-free: it avoids the barren-plateau problem that limits variational quantum models on noisy hardware, which is precisely the regime we will target in Phase 3.

2. QRC Architecture Design

Our architecture follows the realised-volatility QRC of Li et al. [1] and the disordered-ensemble framework of Fujii and Nakajima [2], extended with a two-timescale ensemble readout. It has five components (Fig. 1).

(i) **Reservoir Hamiltonian.** A fully connected transverse-field Ising model on $n = n_1 + n_2$ qubits,

$$H = \sum_{i < j} J_{ij} X_i X_j + v \sum_i Z_i, \quad J_{ij} \sim \mathcal{U}(0, 1) \text{ fixed}, \quad v = 1, \quad (1)$$

with couplings drawn once from a seeded distribution and held fixed. Evolution for time τ is $U = e^{-iH\tau}$. Our $n = 8$ prototype uses $n_1 = 3$ input and $n_2 = 5$ hidden qubits.

(ii) **Input encoding.** At step t the n_1 input qubits are reset and the scaled features are angle-encoded, $R_Y(\pi x_{t,i})$ on qubit i , with features min-max scaled to $[0, 1]$. Angle encoding keeps circuit depth shallow and is hardware-native on gate-based and neutral-atom devices.

(iii) **Memory.** The n_2 hidden qubits are *not* reset. Before each step we trace out the input register, $\rho_{\text{hid}} = \text{Tr}_{\text{in}}[\rho]$, then re-tensor a fresh $|0\rangle^{\otimes n_1}$ and re-encode. This realises a non-Markovian map on the hidden register: information about past inputs persists, while the input register is refreshed deterministically.

(iv) **Readout.** We measure $\langle Z_i \rangle$ on all n qubits, apply a degree-2 polynomial feature expansion, and fit a ridge regression. The polynomial head, shown by Zhu et al. [3] to sharply reduce error, is the only trained component.

(v) **Variants.** The QR2 ensemble concatenates the readouts of two reservoirs at $\tau = 10$ and $\tau/2 = 5$, sampling two memory-nonlinearity operating points; Table 1 reports the better-calibrated single member, QR1 at $\tau = 5$. QR-RU adds *data re-uploading*: within one timestep the encode-evolve block is repeated (here twice, each for $\tau/2$), enriching the input-to-feature Fourier spectrum [11] at fixed total evolution time.

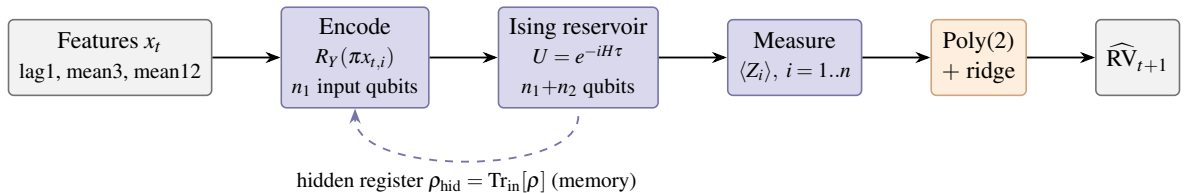


Figure 1: QRC pipeline. Quantum blocks (blue) are fixed and untrained; only the classical polynomial-ridge head (orange) is fitted. The dashed loop is the partial-trace memory channel that carries temporal state between steps.

3. Theoretical and Analytical Justification

Property exploited. The reservoir supplies three resources matched to RV. (1) *Hilbert-space dimensionality*: an n -qubit reservoir lifts the input into a 2^n -dimensional state, and degree-2 readout on n observables

Table 1: Walk-forward out-of-sample performance (101 months). Lower is better for RMSE/QLIKE; MZ slope best near 1; hi-RMSE is RMSE on the top-30% (high-volatility) months. DM is the Diebold–Mariano statistic of QR2 vs each model (negative favours QR2; HLN-corrected p). At $n=8$ the QRC is statistically tied with HAR and a strictly matched ESN, and significantly beats the oversized linear ESN and persistence.

Model	RMSE	QLIKE	MZ slope	MZ R^2	hi-RMSE	DM vs QR2 (p)
QR2 (ens., $n=8$)	0.01738	0.415	0.745	0.208	0.02615	—
QR1 ($\tau=5$, $n=8$)	0.01748	0.406	0.799	0.194	0.02583	−0.24 (0.81)
HAR-OLS	0.01740	0.407	0.666	0.253	0.02567	−0.03 (0.98)
ESN-8 (poly, matched)	0.01740	0.388	0.677	0.241	0.02484	−0.03 (0.98)
QR-RU (re-upload, $n=8$)	0.01786	0.395	0.668	0.213	0.02541	−0.89 (0.38)
ESN-100 (linear)	0.01893	0.473	0.526	0.202	0.02763	−2.15 (0.03)
Persistence	0.01942	0.688	0.490	0.248	0.02939	−1.85 (0.07)

yields $\binom{n+2}{2}$ polynomial features at no training cost. (2) *Dissipation as a design feature*: the partial-trace memory channel is strictly contractive, which Ahmed et al. [4] identify as the mechanism enforcing the echo-state property and generalized synchronization needed to forecast chaotic signals. (3) *A tunable memory-nonlinearity tradeoff*: Čindrak et al. [5] show τ trades reservoir memory against nonlinearity, motivating the $\{\tau, \tau/2\}$ ensemble.

Connection to the signal, made quantitative. We measured the reservoir’s linear memory capacity (MC, Jaeger [12]; for QRC, Fujii-Nakajima [2]) by driving it with i.i.d. input and reconstructing delayed inputs from the *same* degree-2 readout the forecaster uses. At strictly matched readout (8 units, degree 2), the QRC attains MC = 2.92 versus 2.31 for an 8-node ESN, and a higher nonlinear capacity (3.04 vs 2.38); Fig. 2b shows the QRC retains more recent-lag information. The empirical RV autocorrelation has an integrated timescale of 13 months (e-fold 6 months); the reservoir’s intrinsic memory spans the first few lags, so the long-memory component is supplied explicitly by the HAR mean12 feature while the reservoir contributes nonlinear mixing of recent history. MC is stable across $\tau \in [2, 40]$, so the design is not fragile to this hyperparameter. Kornjacá et al. [6] show QRC kernels are geometrically distinct from classical ones, and Hou et al. [7] report a 9-qubit reservoir matching classical reservoirs with thousands of nodes, supporting competitiveness at small n .

Gaps we address. Li et al. [1] establish QRC accuracy on RV but do not characterise noise sensitivity, evaluate under walk-forward, or report matched-readout baselines; no prior work jointly maps encoding density, shot budget and noise for RV forecasting. We begin closing this here and complete it in Phase 3.

3.1 Preliminary prototyping results

We prototyped the full pipeline on S&P 500 monthly RV (1990–2018) under an *expanding-window walk-forward* protocol: the readout is refit before every one-month-ahead forecast, the standard volatility-forecasting evaluation. Table 1 reports 101 out-of-sample months; Fig. 2 shows forecasts, scaling and memory capacity. Reservoir features are computed once and reused across windows.

The honest headline, which is exactly the challenge’s question, is that an 8-qubit QRC is *genuinely competitive* with strong classical baselines: QR2 attains the lowest RMSE of any model, an untrained dynamical map matching a tuned HAR and an equal-readout classical reservoir. At this scale every model is within noise of the others (DM $p \approx 0.98$ for QR2 vs HAR and vs the matched ESN). The matched ESN-8 edges QR2 on QLIKE (0.388 vs 0.415) and high-volatility RMSE (0.0248 vs 0.0262), but both gaps fall inside this $p \approx 0.98$ band and are not statistically meaningful; QR2 significantly beats only the oversized linear ESN-100 ($p = 0.03$) and persistence. The case for the quantum route thus rests not on a significance claim at $n=8$ but on three forward signals: (i) walk-forward RMSE *decreases monotonically* with size, $0.0187 \rightarrow 0.0182 \rightarrow 0.0172$ for $n = 6, 8, 10$ (Fig. 2a, two seeds, single reservoir), the reason we expect the tie to break by $n = 12\text{--}16$; (ii) the matched-readout memory-capacity advantage (Fig. 2b); and (iii) re-uploading improving QLIKE and explained variance. A VIX-derived series through 2026, COVID out-of-sample, confirms the pipeline stays competitive under regime change.

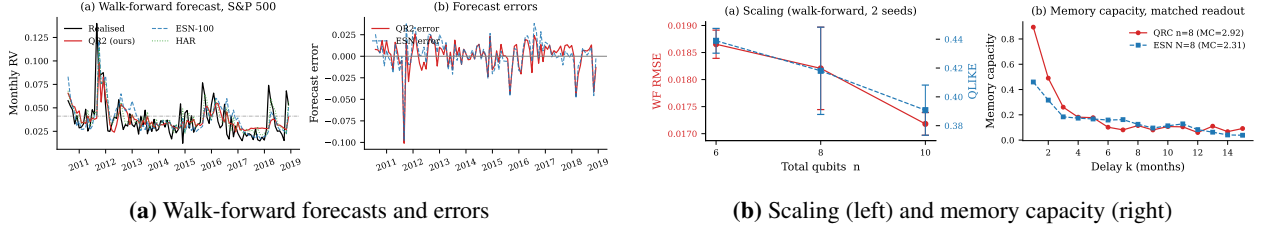


Figure 2: (a) QR2 tracks realised volatility; the dash-dot line marks the high-vol regime threshold. (b) Left: walk-forward RMSE/QLIKE for the single reservoir at $\tau = 10$ fall with size (its $n=8$ point, 0.0182, sits above the QR2 ensemble headline of 0.0174 as expected). Right: at matched readout the QRC retains more recent-lag memory than an ESN.

4. Data Modeling Strategy

Dataset. S&P 500 daily closing prices, 1990–2018 (public historical OHLC; Phase 3 refreshes through 2025 via Yahoo Finance). Monthly RV is $RV_m = \sqrt{\sum_{t \in m} r_t^2}$ from daily log returns r_t , requiring ≥ 15 trading days per month. **Preprocessing.** Core HAR features (1-month lag, 3- and 12-month trailing means), with optional macro features available now (realised quarticity for tails, monthly market return for the leverage effect) and the full macro panel plus forward-wrapper selection deferred to Phase 3. Features are min-max scaled on training statistics for angle encoding; evaluation is expanding-window walk-forward. **Baselines.** Persistence, HAR-OLS, and ESN at matched readout (shown), plus larger ESNs, GARCH(1,1) [8] and an LSTM in Phase 3. ESN is the primary baseline; we report it at strictly matched readout so any QRC gain is attributable to the quantum reservoir. **Metrics.** RMSE, QLIKE [9], Mincer–Zarnowitz, regime-conditional RMSE, and Diebold–Mariano [10], with the Model Confidence Set added in Phase 3.

5. Quantum Platform and Resource Planning

The prototype runs on a noiseless statevector/density-matrix emulation of $n = 6$ –10 qubits. Phase 3 uses, on qBraid: (1) **Aer density-matrix** simulation at $n = 8$ –20 with depolarizing and amplitude-damping channels ($p = 10^{-3}$ – 10^{-2} per layer) for the noise study; (2) a **matrix-product-state** backend with bond dimension $\chi \approx 64$ –256 to reach $n = 20$ –30, validating truncation via the half-chain entropy bound $\log_2 \chi$; and (3) one **gate-based QPU** run with Mitq zero-noise extrapolation. Depth is modest: one encoding layer plus one fixed evolution block per step. The cost driver is the all-to-all coupler: a fully connected U has $\binom{n}{2}$ two-qubit $X_i X_j$ terms, so $n = 12$ needs 66 entangling rotations *if connectivity is native*; on fixed heavy-hex hardware, SWAP routing multiplies depth by $O(n)$, pushing the effective count toward 10^2 – 10^3 . This routing cost, not the bare gate count, sets the platform choice. **Trapped-ion (IonQ Forte)** offers native all-to-all coupling, mapping arbitrary J_{ij} with no SWAP overhead, and is our primary QPU target. **Neutral-atom (QuEra Aquila)**, the analog alternative our citations [6, 7] use, has a fixed van-der-Waals Rydberg interaction rather than arbitrary J_{ij} , so it is a Phase 3 fork contingent on a Rydberg-native reformulation. Shot budgets of 10^3 – 10^4 per observable suffice for $\langle Z_i \rangle$.

6. Stakeholder Impact and Phase 3 Plan

Accurate one-month RV forecasts feed directly into JonesTrading’s core activities: volatility-targeted position sizing, options and variance-swap pricing, and risk-capital allocation, where the regime transitions we target are most costly to miss. The pipeline is hardware-agnostic and open-source, with code released at github.com/Vanshaj0429/artemis-qrc, and the same architecture transfers to Track B weather data by substituting the input features.

Milestones. Phase 2 already delivers the walk-forward protocol, matched-readout ESN, regime metrics, memory-capacity characterisation and re-uploading lever, so Phase 3 builds on validated groundwork. (M1) Extend the scaling curve from $n = 10$ to $n = 12$ –16 on Aer, where the dimensionality argument predicts the matched-baseline tie breaks toward the QRC. (M2) Noise study on Aer density-matrix, testing whether mild dissipation acts as a regulariser [4]. (M3) MPS scaling to $n = 20$ –30, the full macro panel with wrapper selection, and the GARCH/LSTM suite under the Model Confidence Set. (M4) IonQ Forte validation with ZNE, packaged as an agent-executable qBraid Skill.

References

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